

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	PNT2022TMID124356
Project Title	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to predict the Human Development Index (HDI) using Machine Learning techniques. It analyzes socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income per Capita to provide accurate HDI predictions. The solution supports governments, researchers, and organizations in making better development policies and informed decisions.

Project Overview	
Objective	To develop a Machine Learning-based system that predicts the Human Development Index (HDI) using factors such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income. The system provides fast and accurate HDI predictions.
Scope	The project analyzes human development indicators and predicts HDI values using a trained Machine Learning model. It helps users estimate a country's development level efficiently.
Problem Statement	
Description	Calculating the Human Development Index manually is time-consuming and requires analyzing multiple socio-economic indicators. There is a need for an automated system that predicts HDI quickly and accurately using Machine Learning.

Impact	Using this system improves the speed and accuracy of HDI prediction, reduces manual calculations, supports data-driven decision making, and assists researchers, students, and policymakers.
Proposed Solution	
Approach	Develop a Flask web application integrated with a trained Machine Learning model. Users enter the required indicators, and the system predicts the Human Development Index instantly.
Key Features	- Machine Learning-based HDI prediction -Simple and user-friendly web interface -Real-time prediction results



	-Fast and accurate prediction -Easy data input and processing
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 or higher
Memory	RAM	8 GB
Storage	Disk space	1 TB SSD (or 256 GB SSD minimum)
Software		

Frameworks	Python framework	Flask
Libraries	Machine Learning Libraries	scikit-learn, pandas, Numpy, matplotlib, Joblib
Development Environment	IDE	VS Code / Jupyter Notebook
Data		
Data	Source, size, format	Human development Index (HDI) Dataset, CSV Format